

Toward Accurate PAH IR Spectra Prediction: Handling Charge Effects with Classical and Deep Learning Models

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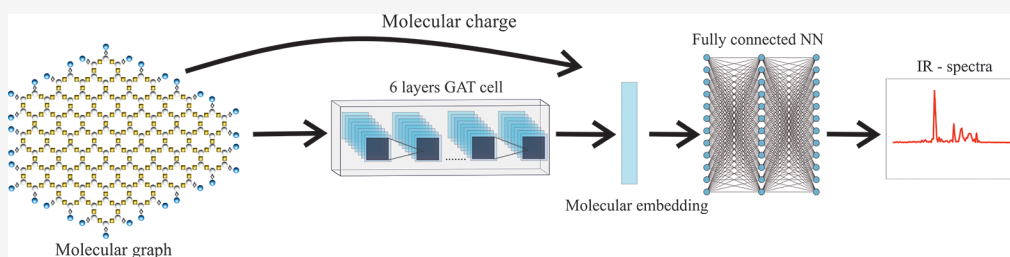
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ABSTRACT: Polycyclic aromatic hydrocarbons (PAHs) play a crucial role in astrochemistry, environmental studies, and combustion chemistry, yet interpreting their infrared (IR) spectra remains challenging due to the similarity of spectral features of many molecules. The presumable presence of both neutral and charged PAHs in mixtures complicates spectra interpretation, too. While first-principle calculations provide accurate spectral predictions, their high computational cost limits scalability. This study employs machine learning (ML) to predict PAH IR spectra, emphasizing the applicability of the developed models simultaneously for neutral and ionized molecules. Two models are introduced: an XGBoost model trained on Morgan fingerprints and a graph neural network (GNN) that employs molecular graph representations. Molecular charges are treated by incorporating their one-hot or learnable NN encodings to molecular representations. Both models demonstrate excellent predictive capabilities, for the first time enabling fast and accurate prediction of charged PAHs IR spectra. While the XGBoost model demonstrates the highest accuracy achieved to date, the GNN shows significant promise for future advancements due to the inherent capabilities of molecular graph representations. Remaining challenges, such as the scarcity of data on heteroatomic PAHs, and potential approaches of addressing them are also discussed in the manuscript.

1. INTRODUCTION

Polycyclic aromatic hydrocarbons (PAHs) and their physicochemical and spectral properties are of significant interest in the fields of astronomy,^{1,2} environmental science,^{3,4} and related fields, including sensor development⁵ and studies on optimization of the combustion process to reduce PAH emission.^{6,7} Environmental studies have focused on PAHs due to their classification as hazardous pollutants, given their toxicity, mutagenicity, and carcinogenicity, which pose significant health risks.⁸ PAHs are widespread in the space, from circumstellar environments to planetary atmospheres and meteorites,^{9–12} making their investigation one of the main objectives in astrochemistry.^{13,14}

Midinfrared (mid-IR) spectroscopy is one of the most widely used techniques for PAH analysis, enabling the characterization of their molecular structure during combustion processes.^{6,7} It is also employed for monitoring the concentration profiles of specific hydrocarbons in air and water.^{5,15} The presence of PAHs in various astronomical objects, including emission, reflection and planetary nebulae,^{16–20} young stellar objects,^{21,22} ultra-luminous infrared galaxies²³ has also been extensively investigated in the mid-IR range. These molecules play a dominant role in heating neutral gas, maintaining the ionization

balance within molecular clouds, influencing star formation,¹³ and contributing to the production of carbon particles and fullerenes.^{24,25}

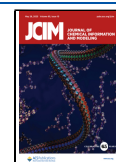
Despite the importance and presumed abundance of PAHs, detailed studies remain limited due to the complexity of interpreting observed IR spectra from astronomical objects, polluted air, or combustion products. These spectra result from the cumulative contributions of tens or even hundreds of PAHs species.^{26,27} Distinguishing signals from different PAHs is particularly challenging due to their spectral similarities. Moreover, the vast number of theoretically possible PAH structures and their various charged states further complicate the IR spectra interpretation, especially in case of astronomical data.¹³ Consequently, the development and application of methods for systematically obtaining IR spectra for as many

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individual PAH molecules as possible are essential for their identification in the environment, interstellar medium, and combustion products.

IR spectra of individual PAHs can be obtained by using both experimental and computational approaches. The experimental measurements of IR spectra for PAHs are limited due to challenges associated primarily with isolating and holding PAHs in the gaseous phase.¹³ In several studies, the infrared multiple photon dissociation (IRMPD) technique has been employed for recording the spectra of small PAH cations.^{28–31} Computational methods are extensively used in combination with such experimental techniques.^{30,32} In particular, density functional theory (DFT), is employed to calculate PAHs IR spectra.³³ However, as the molecular mass increases, DFT calculations become more complicated and resource-intensive.

To accelerate the prediction of the spectra, the application of machine learning (ML) models presents a promising approach. Machine learning has already found successful applications in two domains closely related to the prediction of PAH IR spectra: creating latent representations of molecular structure for further prediction of some properties and the prediction or interpretation of spectral data. Various ML approaches are already used to predict the properties of individual molecules, including solubility,^{34,35} toxicity,^{36,37} melting points,³⁸ as well as for screening molecules for antibacterial activity.³⁹ It is important to note that opinions differ on the relative effectiveness of traditional descriptor-based machine learning methods versus graph neural networks for molecular property prediction. One review suggests that classical models trained on molecular descriptors typically achieve higher accuracy with significantly lower computational cost,⁴⁰ while other highlight several successful applications of graph neural networks in this field.⁴¹ Given these contrasting perspectives, it is reasonable to explore both approaches to address our research objectives. ML has also been successfully applied for predicting the spectral characteristics of various sources,^{42,43} including the prediction of IR-spectra of organic molecules.⁴⁴ Several studies have focused on solving the inverse problem of infrared spectrum interpretation and identifying possible fragments and functional groups in organic molecules.^{45,46} McCarthy and Lee⁴⁷ proposed a method for molecular identification based on experimental rotational data using deep learning networks. Kovács et al.⁴⁸ previously demonstrated the potential of ML for predicting the IR spectra of neutral PAHs using multilayer perceptron (MLP) and random forest model. Despite the success of neural networks to bypass expensive DFT calculations and achieve high-quality predictions for numerous molecules, this approach has excluded ionized molecules from training. More importantly, it prevents the prediction of their spectra, even though they are crucial for identifying emitters and constraining the physical properties of the irradiating source in astrophysical studies.⁴⁹

In this study, we focus on leveraging the significantly expanded PAHs spectral data to enable the prediction of IR spectra for the widest possible range of molecules with a particular emphasis on ionized species. To achieve this, we propose the implementation of two machine learning techniques combined with capacious representations of the molecular structure and ionization state for predicting the IR spectra of PAHs. The first approach employs a classical ML model based on gradient boosting—XGBoost.⁵⁰ The second approach applies a graph neural network.⁵¹ Both methods are capable of providing state-of-the-art results at the moment,

advancing the quality of predicted PAH IR spectra but have different peculiarities that define possible directions of further development.

2. METHODOLOGY

2.1. Database and Data Preprocessing. We gathered data from the publicly available NASA Ames PAH IR Spectroscopic Database (version 3.20).⁵² The chemical structure of the PAH molecules in the database is represented in the form of atomic coordinates (XYZ format). Most of the PAH IR spectra are calculated by using DFT, with only a dozen molecules having experimentally recorded spectra. The low representativeness of experimental data (only 84 unique spectra) prevents models from training solely on them. Another challenge pertains to the noticeable discrepancies between DFT-calculated and experimentally recorded spectra (Figure S1), raising concerns about the validity of their integration into a unified data set. Given that DFT-calculated spectra constitute the majority of the database, totaling 4233 unique spectra, this study uses exclusively theoretical spectra for model training, excluding experimental data from consideration. Since the IR spectra of various molecules consist of a variable number of signals (wavenumbers), we conducted resampling of the spectra to a unified resolution of $\approx 21.33\text{ cm}^{-1}$ across the range $0.21\text{--}5376.69\text{ cm}^{-1}$ through spectral binning. To determine the optimal number of bins, we calculated the average percentage of nonempty bins for each molecule. With 252 bins the average percentage of nonempty bins is approximately 50%. It means that each molecule is well-represented across the spectral range. The resulting spectral resolution of 21.33 cm^{-1} is very close to that used by Kovács et al.⁴⁸ — 21.39 cm^{-1} . We observed that even after spectral binning each spectrum still contains a large gap between 1500 and 3000 cm^{-1} . The low-frequency region of the spectrum contains most of the signals, while the high-frequency part includes only a few. The high-frequency region (from $\approx 3000\text{ cm}^{-1}$) corresponds primarily to stretching vibrations along the C–H bonds or those associated with functional groups and radicals,^{13,52,53} such as $-\text{CH}_3$, $-\text{OH}$, $-\text{CH}_2^\bullet$. Meanwhile, signals in the low-frequency region, up to approximately 1500 cm^{-1} , are primarily attributed to deformation vibrations in aromatic rings. As the gap region contains zero-value points, the low-frequency region (105 bins from 0.21 to 2219.07 cm^{-1}) was selected as the target for machine learning predictions in this study, as it encompasses the majority of spectral signals.

The XYZ format provided in the database⁵² is not straightforward regarding the molecular structure, as it does not contain direct information on the bonds between atoms and other atom properties, but only their coordinates. Therefore, it lacks crucial information for spectra prediction. There are numerous molecular representations that reflect structural formula. One such method is the Simplified Molecular Input Line Entry System (SMILES) representation.⁵⁴ Molecular SMILES strings store information about atom types and properties, bond types, stereochemistry, and aromaticity. Conversion of molecules from XYZ format to SMILES can be performed using the functionality of the RDKit library,⁵⁵ as well as with OpenBabel.⁵⁶ The initial database consists primarily of hydrocarbons but also includes a few PAHs that are dehydrogenated or heterocyclic. While the presence of these molecules does not hinder the format conversion process in general, in some cases, one or the other tool could not handle conversion correctly. Therefore, to transform as many molecules

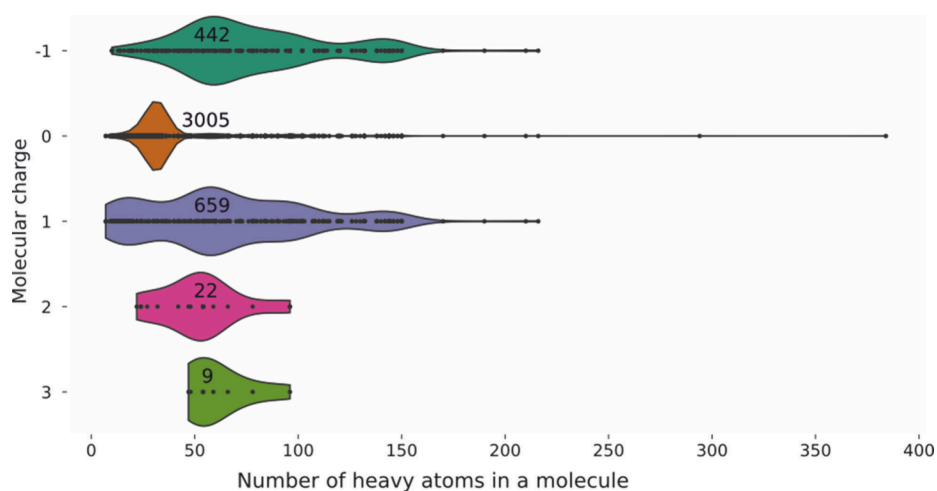


Figure 1. Distribution of molecules by charge and size. The number in each violin plot represents the total number of molecules of the given charge.

as possible from XYZ to SMILES, we used both RDKit and OpenBabel conversion algorithms. The database also contains samples that are complexes of PAHs with magnesium and iron ions, such as Mg^+ , Fe^+ , Mg^{2+} and Fe^{2+} . Since the interaction between the metal ion and the neutral PAH molecule is primarily electrostatic,^{57,58} it is not possible to include such information in SMILES. Therefore, we decided to include such molecules by designating the interaction in these complexes as an “unspecified” bond between the iron or magnesium ion and a carbon atom in a specified ring, depending on the ion’s position in the molecule according to the database⁵² (Figure S2). However, the database still included molecules for which valid SMILES representations could not be generated by any library as well as instances where identical SMILES corresponded to different molecular structures. These molecules were excluded from the data set, resulting in a final processed set consisting of 4137 unique molecules.

2.2. Train Test Split. An important aspect of data in NASA Ames PAH IR Spectroscopic Database is the presence of a notable number of molecular ions, comprising nearly a third of the data set (Figure 1). But the delocalized molecular charge cannot be described by topological descriptors, including SMILES, so the search for an alternative approach for charge encoding was a separate task, which will be discussed in the following sections. Figure 1 also shows a significant decrease in the number of molecules as the number of constituent heavy atoms increases above the values of ≈ 50 –70.

Both distributions across charges and the number of atoms must be considered when partitioning the data into training, validation, and test sets to ensure preservation of the original distributions across all three subsets. To achieve this, we propose a custom train-test split method. First, the data set is divided into quarters based on the number of atoms in molecules. Subsequently, within each quarter, the molecules of each charge are distributed into the training, validation, and test subsets in 70:15:15 ratio (number of molecules: 2898, 620, 619, respectively) for GNN and into the training and test subsets in 85:15 ratio (number of molecules: 3518 and 619, respectively) for XGBoost (test subset is the same for GNN and XGBoost models), ensuring that the original charge state distribution is preserved in each subset.

2.3. Molecular Descriptors and Models. **2.3.1. Classical Machine Learning Approach.** Prior to any learning step, SMILES representations of molecules should be converted into

a numerical format suitable for input into a model. Usually, molecular descriptors are employed as numerical representations of a molecule’s structural and chemical characteristics in tasks of similarity search or ML prediction of properties. These descriptors encode comprehensive information about the types and properties of constituent atoms, their local environments, and bonding patterns. Among these methods are molecular fingerprints, which result from fragmenting molecules and encoding these fragments into bit vectors. The position of a number in a molecular fingerprint corresponds to a structural fragment, while the value indicates the number of such fragments or their absence. In this work, we use the Morgan Count Bit Vector (Figure 2), where bits represent the count of

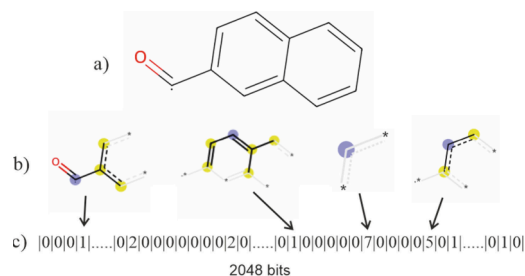


Figure 2. Molecular fragmentation into a bit vector: (a) structure of the molecule; (b) molecular fragments; (c) Morgan count bit vector.

each fragment in molecule formed by Morgan’s algorithm.⁵⁹ The number of bits is a variable parameter that can be set directly, enabling precise control over the fingerprint’s resolution and sensitivity in representing molecular features. This approach also avoids the challenges observed earlier⁴⁸ with the molecular fingerprint generation, when the number of features grows rapidly and becomes difficult to control as the data set expands. We set the number of bits for the Morgan Count Bit Vector equal to 2048, as this number of fragments provides detailed enough representation of the structure of each molecule in the data set (Figure S3).

To enable the distinction between molecular representations with varying charges, we used a one-hot encoding method. The charge of each molecule was converted into a 5-bit vector that was concatenated with the corresponding molecular fingerprint. Thus, each molecule is represented by a unique feature vector of length 2053.

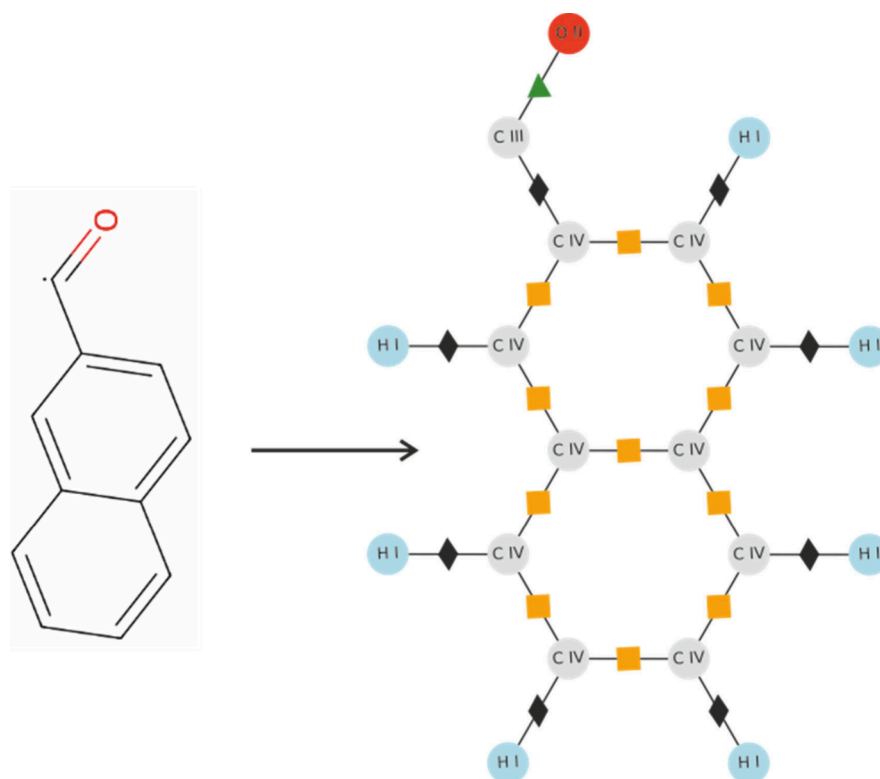


Figure 3. Molecular graph representation. The different bond types are marked as follows: yellow square, aromatic; black diamonds, single; green triangle, double.

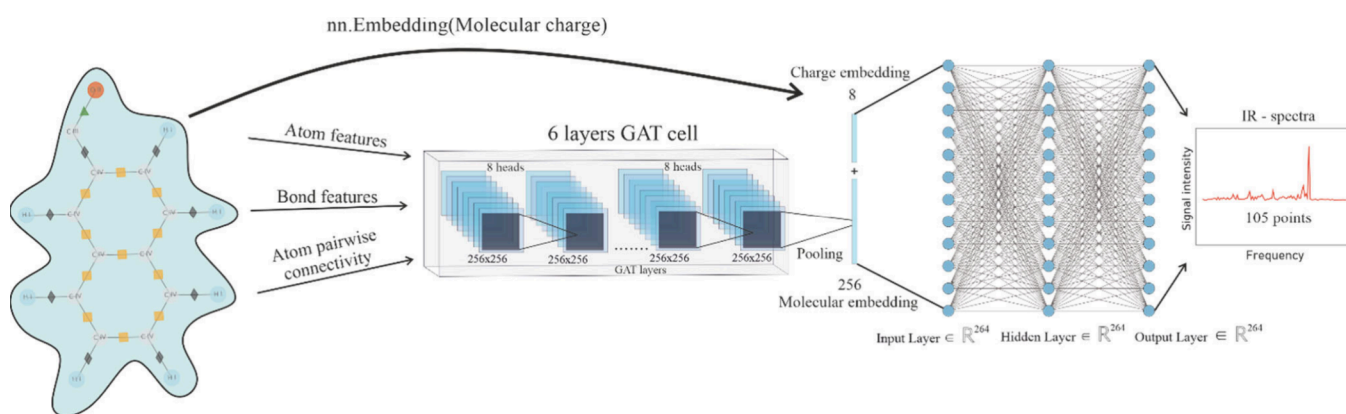


Figure 4. Graph neural network architecture.

After the data transformation, the molecular fingerprints were fed into the XGBoost model to predict the IR spectra. Hyperparameters optimization of the XGBoost model was done with the open-access Optuna library⁶⁰ with TPESampler and 5-fold cross-validation to ensure resource efficiency, model robustness and to prevent overfitting. The ranges of hyperparameter optimization, as well as the optimal values, are given in (Table S1).

2.3.2. Graph Neural Network (GNN). Every molecule can be represented as a graph, defined as a set of nodes corresponding to the atoms in the molecule and edges indicating the connections between nodes, thus reflecting the chemical bonds (Figure 3).

Graph neural networks (GNNs) are employed for interacting with molecular graphs and extracting valuable information. These networks rely on operations such as graph convolution

and message passing. GNNs receive vector representations of nodes and edges, as well as an adjacency matrix or pairwise connection tensors, which are encoded from atom and bond properties, such as atom type, bond type, charge, hybridization, etc. The message-passing mechanism involves the exchange of information between adjacent nodes. Each atom is assigned an embedding vector that incorporates both its own features and the embeddings of the atoms to which it is chemically bonded. Through the convolution operation, messages are aggregated and averaged, leading to an update of atom embeddings at each convolutional layer. Additionally, bond embeddings, which facilitate message transmission between atoms, may also be included.^{51,61}

The architecture of the graph neural network developed in this study is composed of two primary components: a graph block and a fully connected block. The graph block consists of

graph convolution layers enhanced with an attention mechanism,⁶² which computes weight coefficients for each neighboring atom. The Graph Attention Network (GAT) layer convolutions allow for varying contributions from each atom during the update of feature embeddings, thereby enabling the model to capture the differential influence of atoms within the molecular structure more effectively. For designing the GNN architecture, we employed the PyTorch Geometric library,⁶³ which provides a comprehensive set of tools for working with graph-structured data. At the output of the graph convolution block, the obtained node feature embeddings of each molecule are aggregated and pooled to form molecular (graph) fingerprints. Subsequently, the embeddings are processed through fully connected layers, resulting in the predicted IR spectra at the output. The described architecture of the graph neural network is illustrated in Figure 4.

As well as in the case of classical ML models described above, the process of learning molecular embeddings does not account for molecular charge, as graph representations are purely topological mappings of molecules. To encode the charges, we employed the learnable embeddings from the PyTorch library,⁶⁴ which generate vector representations that are updated during the model's training process. These embeddings are concatenated with the molecular embeddings following the pooling stage. Optimization of GNN parameters, including the number of graph and linear layers, channel dimensions, and number of heads in the GAT cell, was also performed using Optuna.

2.4. Loss Function and Metrics. Considering the complex structure of the targets (IR spectra), where both peak intensity and position are crucial and values within a peak are inherently correlated, the choice of loss function is particularly important. It must ensure that the optimization process produces physically meaningful results while effectively minimizing the value of loss function. When choosing a loss function for the prediction of IR spectra of PAH molecules, it is essential to consider two major features: the accuracy of the predicted signal position and the spectral shape, determined by the relative signal intensities and signal widths. Additionally, given the relatively small data set size (4137 molecules), it is advisable to select loss functions that are robust to outliers and provide strong generalization capabilities. Classical loss functions, such as Mean Absolute Error (MAE) and Mean Squared Error (MSE), while easy to interpret, are not well-suited under these conditions. MSE is sensitive to outliers, whereas MAE may impose insufficient penalties on local deviations. Therefore, we prefer the Huber Loss function. It serves as a compromise between MAE and MSE: it behaves quadratically for small deviations and linearly for larger ones. This balance allows it to maintain robustness to outliers while simultaneously focusing on significant errors. The Huber Loss function is defined as follows:

$$L_{\delta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & \text{if } |y - \hat{y}| \leq \delta \\ \delta|y - \hat{y}| - \frac{1}{2}\delta, & \text{otherwise } |y - \hat{y}| > \delta \end{cases}$$

where y is the target, $\hat{y}$ is the predicted value, and δ is a hyperparameter that sets the threshold at which the loss function transitions from quadratic to linear. Another function that combines quadratic and linear behavior is the Pseudo-Huber Loss:

$$L_{\delta}(y, \hat{y}) = \delta^2 \left(\sqrt{1 + \left(\frac{y - \hat{y}}{\delta} \right)^2} - 1 \right)$$

Unlike Huber Loss, Pseudo-Huber Loss is a smooth function, providing a more gradual transition between the MSE and MAE.

Another important aspect is the selection of a metric that most accurately captures the difference between the predicted and true values. If we consider the IR spectra represented as intensity vectors, then the cosine similarity metric can be used to calculate the difference between them. It is computed as $\cos(A, B) = \frac{A \bullet B}{\|A\| \|B\|}$, where $A \bullet B$ is the dot product of vectors and $\|A\|$ and $\|B\|$ are the norms of vectors A and B , respectively. Cosine similarity measures the angular convergence of vectors based on their orientation. For spectra represented by vectors with non-negative intensity values, cosine similarity ranges from 0 to 1, where 0 indicates that the vectors are orthogonal (the least similarity), and 1 means the vectors have the same direction (identical spectra). The advantage of cosine similarity lies in its sensitivity to signal position and relative intensities within the spectra. It does not account for differences in absolute intensity values between two spectra, which is particularly useful when the spectra are not normalized to a single scale. Implementation of cosine similarity as a loss function in PyTorch⁶⁵ is called Cosine Embedding Loss.

Another metric for ML spectra prediction applied by Kovács et al.,⁴⁸ is the Earth Mover's Distance (EMD), or alternatively the Wasserstein metric, evaluates the difference between two distributions (spectra) by defining "distance" as a measure of their dissimilarity. These distributions are thought of as histograms, and EMD is interpreted as the minimum amount of work needed to redistribute values across bins until the two histograms are fully aligned. A lower EMD value indicates a higher similarity between the distributions. Mathematically, it is expressed as follows:

$$\text{EMD}(a, b) = \sum_i \left| \sum_{j \leq i} (a_j - b_j) \right|$$

where a_j and b_j are the j -th elements of distributions a and b , respectively. Applied to spectra, EMD ensures that peak positions, including their relative alignment, are considered. This aspect can support a more accurate reconstruction of the predicted spectral shapes. However, the absolute values of spectra intensity also contribute to EMD, making it essential to normalize the spectra beforehand to ensure optimal algorithm performance.

3. RESULTS

3.1. XGBoost and GNN Predictions. All of the spectra predicted by XGBoost discussed in this section were obtained using the optimized model trained with the built-in Pseudo-Huber loss function. For a fair comparison of classic ML and GNN models as well as for GNN optimization, we required the same metric. To determine the most suitable metric, we trained two instances of GNNs, optimized with EMD loss and Cosine Embedding Loss. Rather than focusing on their absolute predictive accuracy, we thoroughly analyzed patterns in their predictions to identify the most appropriate metric.

Without delving into the numerical or statistical analysis of cosine similarity and EMD values, we observed that using EMD

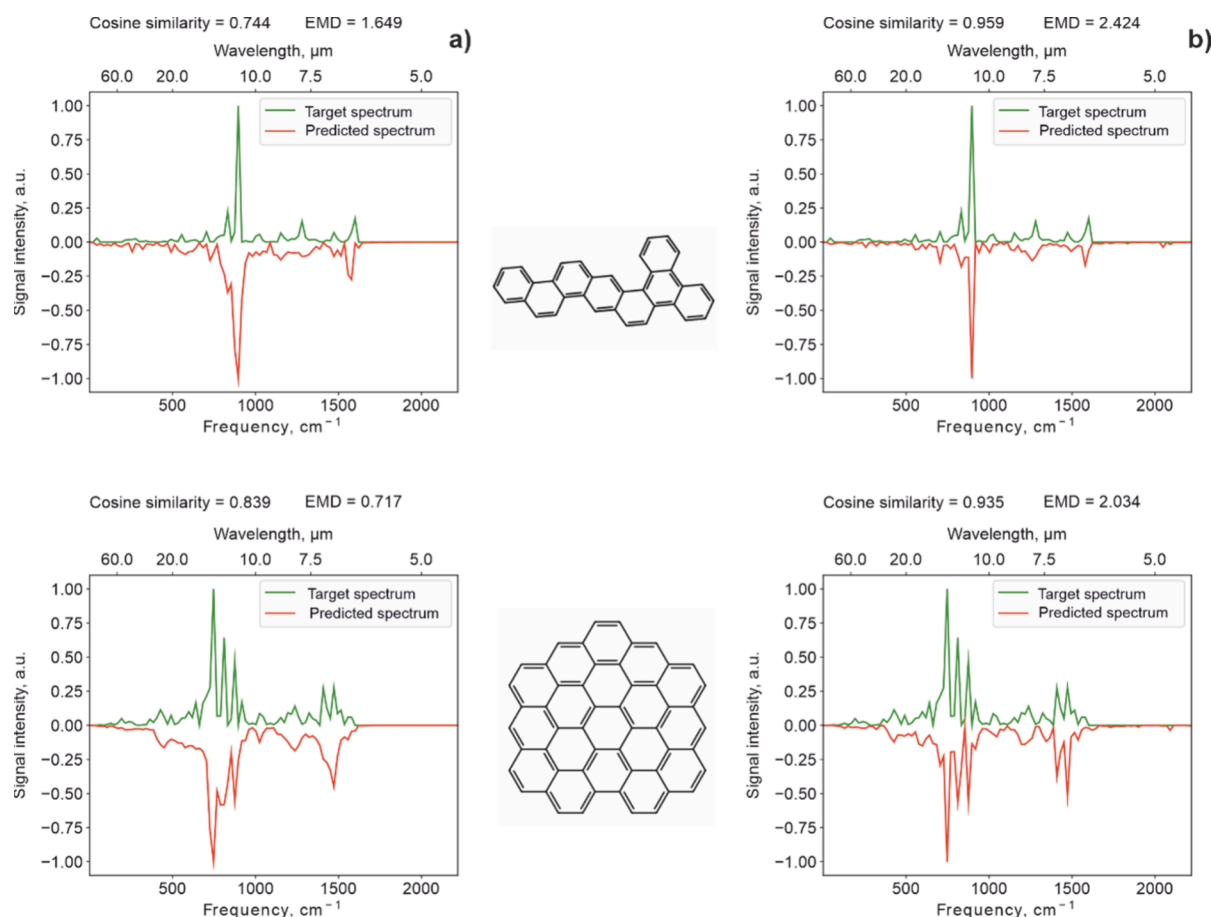


Figure 5. Predictions of PAH IR spectra using GNN trained with (a) Earth Mover's Distance (EMD) and (b) Cosine Embedding Loss as the loss function.

loss (Figure 5a) resulted in broader and less aligned predicted signals (red curves) compared with the true signals (green curves), particularly the most intense peaks. In contrast, training the model with Cosine Embedding Loss (Figure 5b) led to predictions that exhibited a closer alignment in both the shape and the intensities of the target spectra. Absolute values of cosine similarity and EMD for cases *a* and *b* (Figure 5) show that cosine similarity metrics correlate better with the general ideas on the quality of the predictions (match in peak position and signal shape with the target spectrum). For example, for the top-row molecule, cosine similarity indicates that model *b* prediction is more accurate (0.96 vs 0.74), while the EMD metric leads to opposite: EMD of 1.6 and 2.4 in the case of *a* and *b*, respectively. But the discrepancies between the predicted and the target spectra in case of model *a* are more obvious: prediction exhibits worse resolution between major peaks (800–1000 cm^{-1}), nonzero background from ≈ 1100 to 1500 cm^{-1} and broader signals in general. The latter is of major importance since the overall spectral resolution (21.33 cm^{-1}) is already far from ideal due to a lack of data. Therefore, we chose cosine similarity as the metric function for all further studies, and we consider that it correctly indicates the quality of predicted spectra.

The average values of cosine similarity on a common test set for all models are summarized in Table 1. The mean metric values reveal an expected trend for a group of GNNs, where the highest average cosine similarity is observed in the predictions made by the model trained by using the Cosine Embedding Loss function. However, while it shows the best average prediction of

Table 1. Predictive Performance of the XGBoost and GNN Models

model (loss function)	cosine similarity
XGBoost (Pseudo-Huber)	0.789
GNN (Cosine Embedding Loss)	0.764
GNN (Huber loss)	0.740
GNN (EMD)	0.717

all GNNs, the model trained with the Huber Loss excels in individual prediction accuracy, achieving the highest cosine similarity score for a single molecule (0.995 vs 0.992 for the one with Cosine Embedding Loss) and also mitigates the worst-case error, giving the worst value of cosine similarity of 0.115 opposed to 0.049 for the model with Cosine Embedding Loss. Despite these notable examples of the edge-cases when training the GNN with Huber Loss, it is evident that Cosine Embedding Loss is more suitable for predicting the IR spectra of PAHs due to its overall quality and consistency of results. The closeness of the results achieved by the XGBoost and GNN models (Table 1) demonstrates their ability to predict spectra with high accuracy. However, deeper insight into the nature of errors of both classes is required for a recommendation of the best model.

To demonstrate the impact of charge on the spectra and to enable a comparison between the developed models as well as with the results of Kovács et al. study,⁴⁸ we trained the models without inclusion of charge representation. This led to moderate prediction accuracy (Table 2), with average cosine similarity values of 0.694 and 0.704 for XGBoost and GNN, respectively.

Table 2. Predictive Performance of Models Based Solely on Structural Representations of the PAH Molecules

molecular charge	cosine similarity	
	XGBoost	GNN
−1	0.742	0.705
0	0.707	0.716
+1	0.693	0.654
+2	0.634	0.511
+3	0.673	0.705
average	0.694	0.704

These findings highlight the necessity of explicitly incorporating molecular charges into molecular representations.

As a next step, we carefully considered the variability of molecules across their size and charge while defining encoding and splitting strategies; therefore, the assessment of their efficacy is needed. We analyze IR spectra of structurally identical PAHs in different charge states from the test set to assess the models' ability to differentiate between charged and neutral molecules, identify their characteristic spectral features, and uncover patterns in the generated output. Figure 6 illustrates examples of such molecules and the corresponding IR target (green) and predicted (red) by XGBoost and GNN spectra. These molecules are present in the test set in neutral, negatively, and positively charged forms. As distinct structural units, the three of them differ solely in size (basically, the number of aromatic rings) and share identical structural properties, consisting only of carbon and hydrogen atoms without any functional groups.

The target IR spectra of molecules with varying charges show notable differences. The presented spectra of PAHs with a charge of (−1) contain two prominent signals near 1200 and 1500 cm^{-1} , along with a smaller signal around 700 cm^{-1} . The one at 1200 cm^{-1} has notable “splitting”. For positively charged PAHs (+1), the spectral features resemble those observed in the case of a negative charge. However, the signals are slightly shifted. Notably, the signal near 1200 cm^{-1} is less split compared to the (−1) charge state. Another trend is observed with changes in PAH size: as the molecule size increases, the signals in the 1200 cm^{-1} region for cationic and anionic PAHs broaden and merge into a single peak. In the case of neutral molecules, the spectral pattern differs significantly. A strong, singular signal is present near 700 cm^{-1} , with all other peaks being comparatively weaker. A much deeper discussion of assignment of spectral features across the IR-range can be found elsewhere,⁵³ but we focus primarily on the models' capabilities to follow major trends. We can state that both the XGBoost and GNN models, trained on the corresponding representations, successfully recognize key features that allow them to distinguish between molecular charges, as clearly demonstrated in Figure 6. The high values of cosine similarity confirm the effectiveness of both charge encoding approaches. The GNN performs slightly better in predicting the shapes of signals of charged molecules, for instance, more accurately depicting the “splitting” of the 1200 cm^{-1} signal. This fact may be attributed to the enhanced representational capacity of graph embeddings, characterized by their higher dimensionality and learnable embeddings for molecular charge. At the same time, XGBoost provides more accurate predictions in the region of high wavenumbers (around 2000 cm^{-1}), where no significant signals are present for these example molecules. In this region, the predictions of the GNN for charged molecules are noisy and exhibit small fluctuations. The models' predictions also reflect differences in molecular

size. The predicted spectral shapes and relative intensities differ distinctly with the change of molecular size and follow the same descriptive trends as those given for the target spectra.

The mean cosine similarity on the test set across different charge states (Table 3) demonstrates a decline as the molecular charge increases from (0) to (+2). This trend aligns with the representativity of molecules in the data set, as illustrated in Figure 1. The decrease in the total number of molecules in the train set with increasing charge is expected to diminish the models' learning capacity. However, the highest mean cosine similarity values are observed for molecules with charges of (−1) and (+3). Based on a comparison of Table 2 and Table 3, we can conclude that the suggested approaches, based on one-hot encoding and learnable embeddings, demonstrate their effectiveness in capturing charge-related properties.

A thorough consideration of the results revealed one possible reason for the low prediction accuracy cases. The accuracy is influenced by the presence of heteroatomic molecules. It gets worse when heteroatomic PAHs are included, although they remain under-represented, and the metric shows higher values when all molecules in the subset (corresponding to a single charge state) exhibit greater structural uniformity. The IR spectra of heteroatomic PAHs (e.g., those with O or N in aromatic rings) can vary significantly from those of ordinary PAHs, and since their representativity in the data set is extremely low, the prediction results tend to deteriorate considerably when such molecules are present in the test set. The number of neutral molecules containing nitrogen or oxygen atoms within aromatic rings or in functional groups in the test set is 9, compared to 17 for positively charged molecules (+1) and only 1 for negatively charged molecules (−1). Notably, the highest metric value across these three charge states is observed for the negatively charged molecules (−1), while the singly positively charged molecules (+1) exhibit the lowest metric value among these three charge states. The high cosine similarity observed for the (+3) charge state can be attributed to its limited representation in the data set, with only one molecule in the test set and six in the training set, alongside the uniformity of molecules, as indicated by the absence of heteroatomic molecules among the (+3) PAHs. More discussion of predicting spectra of heteroatomic molecules will follow in Section 3.2.

3.2. The Past, Present, and Future of PAH IR Spectra Predictions. First, and most importantly, we introduce a harmonious expansion of the concept of ML-based prediction of PAH IR spectra by addressing, for the first time, the problem of predicting spectra across different charge states of PAHs. Second, our approach to molecular representations – using Morgan Count Bit Vector for XGBoost – offers a more universal and efficient alternative to the previously used Morgan Fingerprints for PAH IR-spectra prediction.⁴⁸ This representation provides a reduced dimensionality that can be controlled directly, enhancing its versatility. Finally, to ensure a statistically based comparison of the prediction quality we trained optimal instances of our XGBoost and GNN models, as well as the fully connected network proposed by Kovács et al.,⁴⁸ and tested their performance on a unified test set composed exclusively of neutral PAH molecules. In the latter case, charged PAHs were excluded from the test set as the model was not designed to account for molecular charge. The training set described in Section 2.2 was used for our models. The distributions of cosine similarity values on the test set are illustrated in Figure 7.

The higher mean cosine similarity values for both the XGBoost and GNN models than for fully connected neural

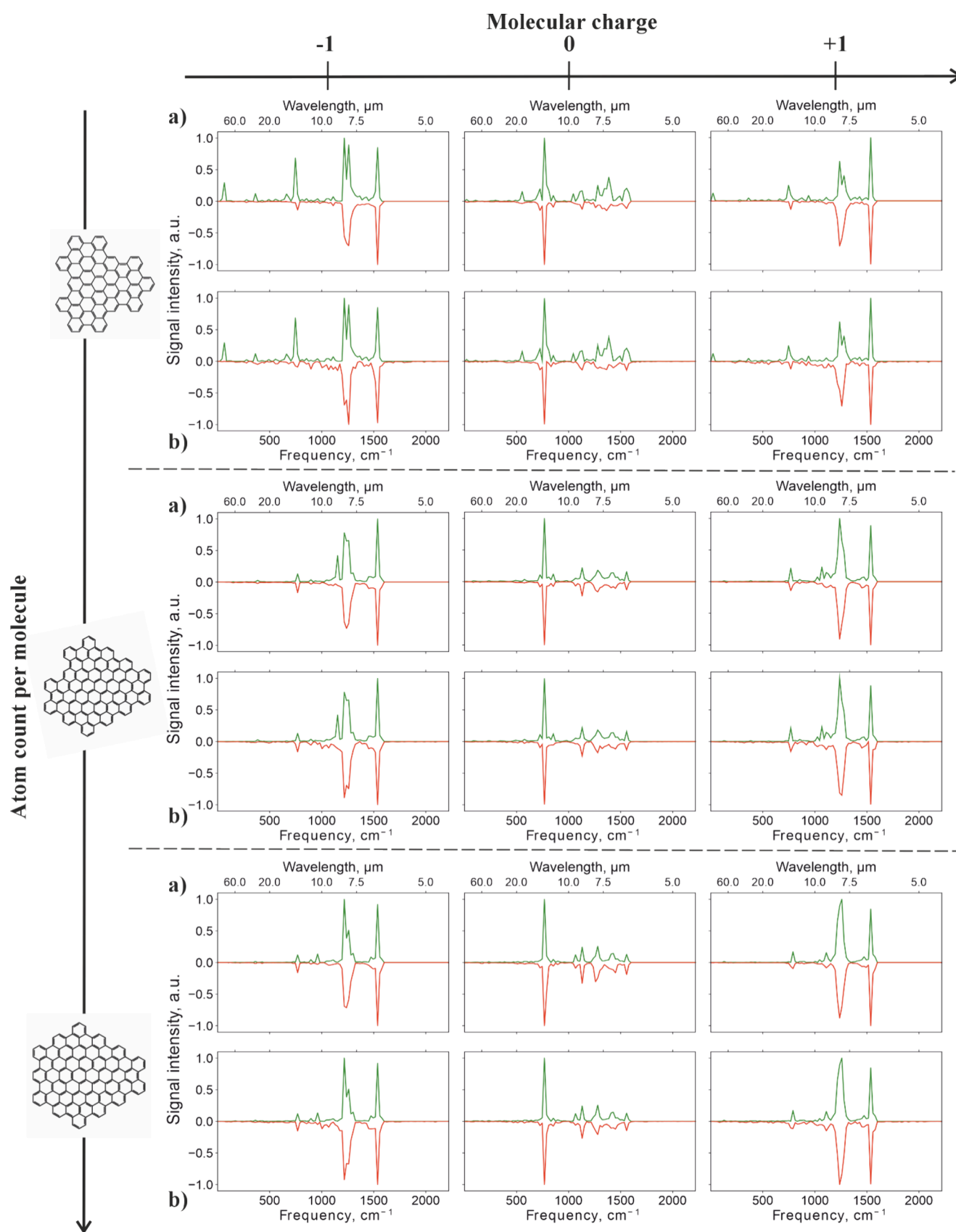


Figure 6. Prediction accuracy of IR spectra for molecules with different charges: the green line represents the target spectra and the red line indicates the predicted spectra: (a) XGBoost predictions; (b) GNN predictions.

network indicate the obvious benefits of suggested molecular and charge encodings as well as the model architecture in terms

of average precision of neutral PAH IR spectra prediction. Analysis of quartile ranges of these distributions provides some

Table 3. Mean Metric Value for Each Charge State

molecular charge	cosine similarity	
	XGBoost	GNN
−1	0.800	0.801
0	0.794	0.766
+1	0.763	0.739
+2	0.736	0.541
+3	0.824	0.847
average	0.789	0.764

additional insights. The maximum achievable cosine similarity for a single molecule from the test set by the XGBoost and GNN models (0.992 and 0.989, respectively) exceeds that of the fully connected neural network (0.971). The maxima of probability density function (PDF) values are also located at higher metric values in the case of both of our models. A similar trend is observed at the third quartile and median value ranges. However, the shape of the PDF distribution remains nearly the same across all three models. The situation differs in the lowest quarter, where most molecules from the under-represented class of heteroatomic molecules are found. This range exhibits a relatively flat distribution with the highest Q1 (25% boundary) value for the XGBoost model, while the worst-case cosine similarity is comparable across all models, with a slight advantage for the fully connected neural network. Indeed, upon examination of the molecules with the lowest accuracy, a significant proportion of PAHs containing heteroatoms, either within the aromatic ring or as functional groups, is observed. Examples of structures and spectra of such molecules are presented in Figure 8.

Moreover, the group of PAHs with the poorest prediction accuracy includes dehydrogenated molecules composed solely of carbon atoms. Since there is a limited number of heteroatomic PAHs in the data set, to tackle the challenge of low prediction accuracy of the IR spectra of heteroatomic PAH molecules, we assumed that a transfer learning approach can be employed in future study. It would involve implementing a method commonly used in neural networks domain, where a pretrained on a large data set model is further fine-tuned on a smaller but more specific. If one can select molecules structurally similar to PAHs, including those containing nitrogen and oxygen atoms, and apply a transfer learning technique, it might be possible to enhance IR spectrum predictions not only for heteroatomic PAH molecules but also for all PAHs in general. The primary task is to establish a criterion for evaluating a molecule's suitability, based on its structural similarity to PAHs, for the proposed approach. Additionally, the molecule's IR spectrum should be accessible. As mentioned earlier, the high-frequency region of the spectrum features signals influenced by stretching vibrations within functional groups (e.g., NH_2 , OH). Including this spectral region may improve the model's predictive accuracy for heteroatomic PAHs. Thus, we have identified the primary limitations of our models and outlined the possible future direction for enhancing PAHs IR-spectra prediction.

4. CONCLUSIONS

We have demonstrated the predictive power of both classical ML approaches and graph neural networks trained on the largest data set of DFT-calculated IR spectra of PAH molecules available to date. The XGBoost model currently represents the most effective approach for spectral prediction tasks, as evidenced by its consistently superior performance on average

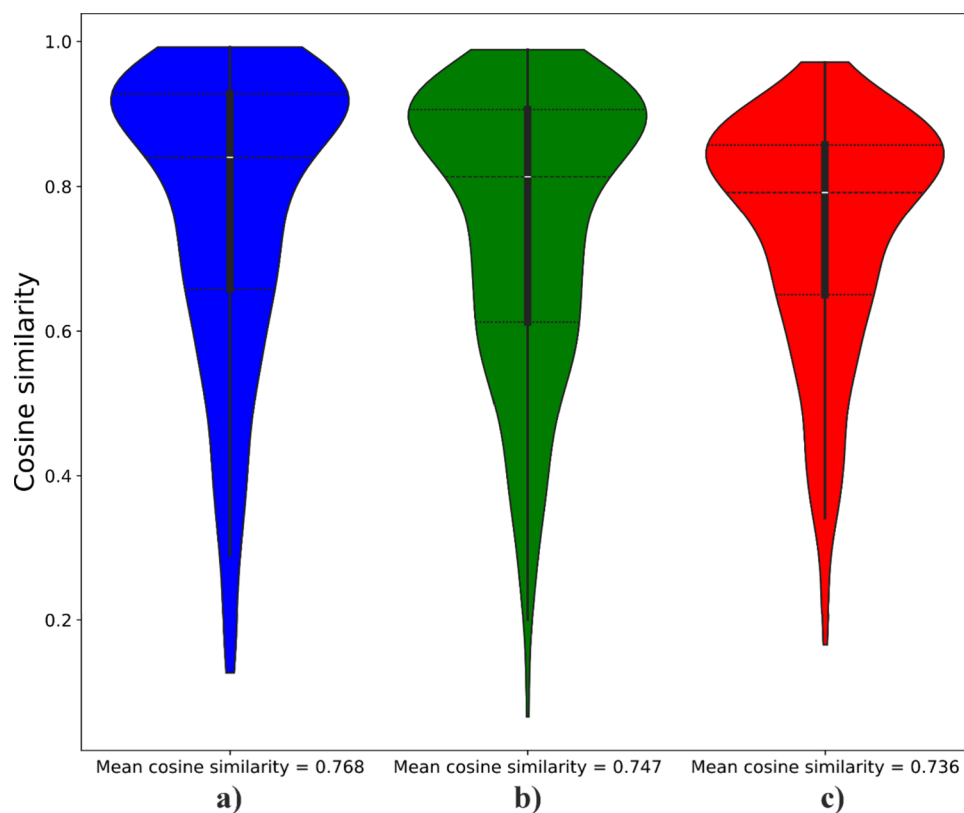


Figure 7. PAH IR spectra prediction accuracy by models: (a) XGBoost; (b) GNN; (c) Fully Connected Neural Network proposed by Kovács et al.⁴⁸

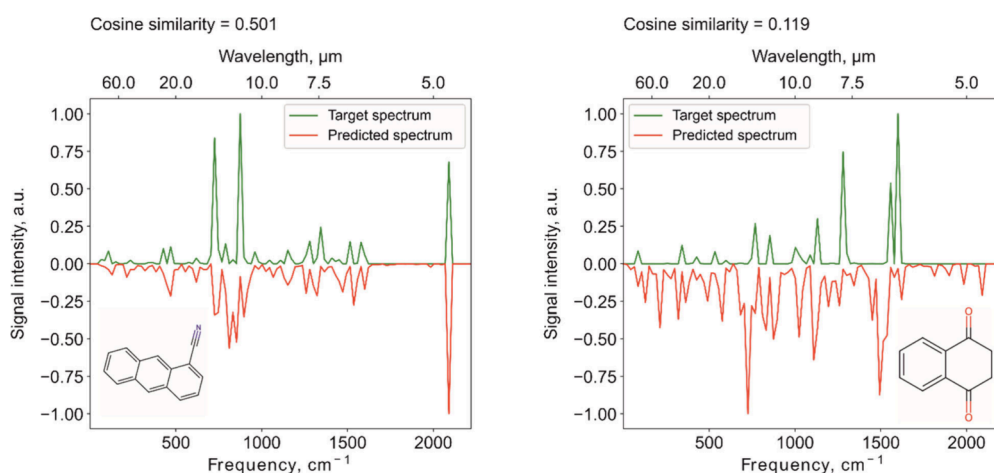


Figure 8. Predicted IR spectra by GNN for heteroatomic PAH molecules.

and, in extreme cases, compared to prior methodologies and the GNN model.

Despite this, the GNN model shows significant promise for future advancements. Its potential lies in its inherently flexible and expressive molecular graph encoding framework, which facilitates detailed and nuanced representations of molecular structures. The GNN's ability to incorporate transfer learning provides an opportunity to further improve its predictive performance and broaden its applicability.

This study marks a step forward by enabling the prediction of PAH ion spectra through the comprehensive encoding of molecular charge states. Accurate and detailed encoding has been shown to enhance the versatility and applicability of predictive models considerably. The achieved accuracy—both averaged across the independent test set and within specific charge states and molecular size categories, enables reliable prediction of spectra for a wide range of PAH structures. These advancements pave the way for the precise modeling of PAH mixture spectra, addressing the critical challenge of investigating the compositional properties of astronomical objects.

■ ASSOCIATED CONTENT

Data Availability Statement

The data set, models and code used to generate the results in this paper are available on Zenodo.⁶⁶ The data set files are also provided as [Supporting Information](#).

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c00372>.

Supplementary figures and tables referenced in the text ([PDF](#))

A database of PAH molecules containing UIDs, SMILES, and additional data (database.csv); The test set used to evaluate the prediction accuracy of PAH IR spectra by the XGBoost and GNN models (database_test.csv) ([ZIP](#))

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Author Contributions

The manuscript was prepared through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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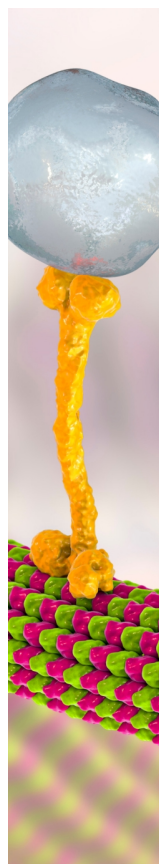
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